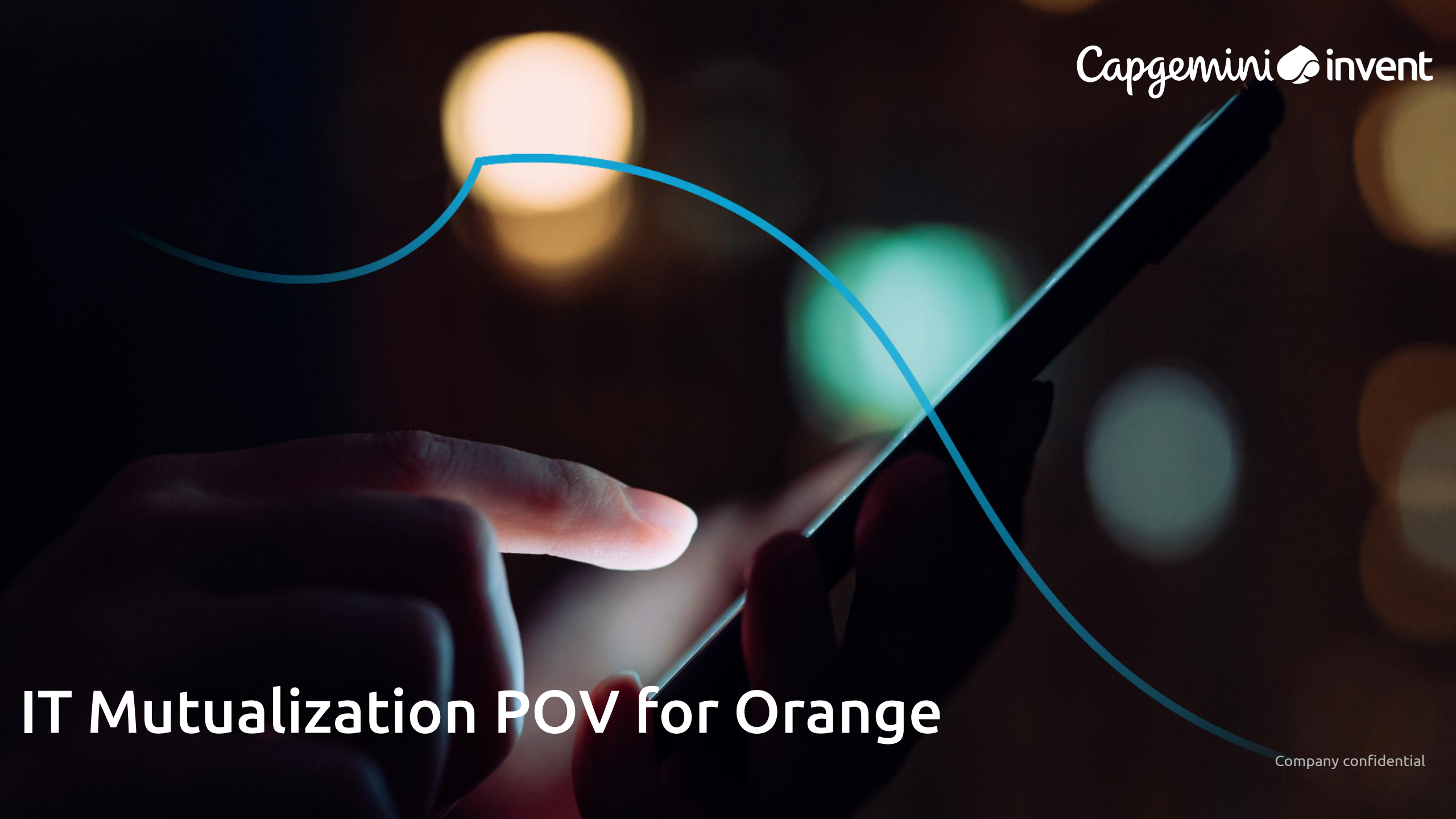


IT Mutualization POV for Orange

Company confidential

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Background/Summary

BSS (Business Support Systems) IT Mutualization refers to consolidation, sharing, and co-ownership of core telecom IT platforms. TM Forum Open Digital Architecture (ODA) enables IT mutualization by decomposing end-to-end business capabilities into standardized, loosely coupled components with open APIs, allowing shared platforms and services to be reused consistently across multiple business units, brands, and channels while preserving local autonomy.

WHAT

- Implement **Single-Architecture, Multi-Country**
- **Unified Product Catalog** Multi-country Model
- **Standardized Operating Model**
- **Shared infrastructure**, Faster Upgrades, Easier Maintenance
- **Accelerate Time to Market**
- **Improve data quality**, decision making with **consistent KPI's**
- Investments **focused on Value Delivery** and **Customer Experience**

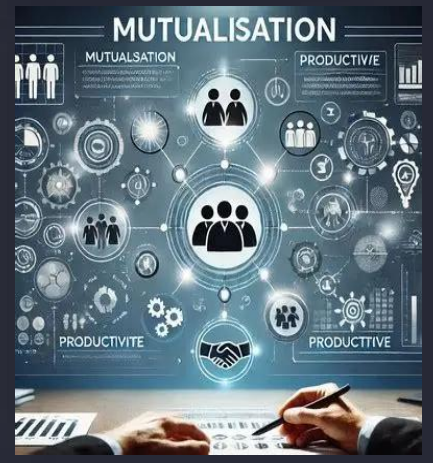
WHY

- **Average tier-1 telco spends 15–25%** of its IT budget **on BSS**
- Support a strategic and **digital-native transformation**
- Scale to **fast-changing B2B digital services** landscape
- **Enhance global support operations**
- **SLA based contracts**: Service level, Customer-level or Customized
- **Improved Cost Efficiency**. Savings of 20–40% on BSS TCO are commonly cited in TM Forum benchmarks
- **Centralized security** architecture and control
- **Unified data governance**



HOW

- **Wide-scale transformation**, **executed in a phased manner**
- **Adopt ODA Architecture Principles**, Utilize Open APIs
- Mutualized **Cloud-Native BSS Microservices** Platform
- A **step-by-step approach** towards **rationalizing the application landscape**
- A **clear migration strategy** alongside a **defined operating model**
- **Multi-language billing** and **customer services**
- **Well defined operating-model** change and cultural adoption
- Agreed approach towards **Governance mutualization** across Product, Architecture, Business, IT and Operations
- **Standardized and platform based global solution** across business lines & processes



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Benchmark View for Mutualization | Business Unit Costs vs Market Entry Efficiency

1global Benchmark View

A Architecture

- Cloud native & Centralized Architecture
- Platform 1st Model across markets

1 BSS platform

- Highly Federated structure design with market driven customizations
- Layered structure, Multiple logics

L

B Multiple Tenancy Structure

- 1 Application –Multiple markets /customers
- 1 CRM & Provisioning Application across Markets

12 countries

- Limited tenancy with coexistence design
- Higher costs to serve per customer /market

L

C Catalog Design

- Unified Global Catalog. Market specific differentiation are minimal
- Minimal Child Structures

95% Reusability

- Complicated Catalog design. Complex parent-child linkages and non optimized catalog

M

D Charging Mechanism

- Non Market specific charging mechanism. Well defined digital channels

95% standardized

- Standardized Real time charging but still missing cloud native convergent charging platform

M

E API & Configuration Design

- API driven expansion to new markets & subsequent monetization
- Open Industry specific APIs

API monetization

- Highly localized non-standard APIs not standard across B2B, B2C, B2B2C

L

F Cost Control

- Marginal Incremental costs outlay due replication & reuse philosophy
- Minimal market entry costs

50%-60% less costs

- High Costs incurred due to localized deployments
- High O&M Costs

L

Typical Operator Benchmark View

Data collected from industry reports/open sources

Leader position

Position relative to industry best performer

1global digital platform

Needed a real-time BSS that supports flexible multi-play commercial models for existing and new business lines, agile rollout of new functionalities and features derived from a complete end-to-end CI/CD pipeline to react quickly to market needs and auto scale capabilities

Cerillion Product Catalogue (2013)

- Catalog-driven convergent BSS model
- Tightly integrated
  catalog ↔ billing ↔ charging
- Single product catalogue to support multi-country business model.
- Enabled rapid configuration for new markets (config driven product definition)
- One global catalog instance (Country behaviour driven by attributes & rules, not duplicate products)

Optiva Charging Engine (2022)

- Implemented Optiva Charging Engine (OCS) on the public cloud with cloud native, open architecture for its business support ecosystem (a real-time converged charging solution that is 5G ready)
- Optiva Charging Engine runs on Google Cloud Platform (GCP) that uses Cloud Spanner.
- Cloud Spanner provides strongly consistent transactions and scales horizontally to thousands of nodes across regions
- Optima Charging is the real-time online charging system that controls and authorizes live network usage at session speed

Cerillion Convergent Billing & CRM (2013)

- B2C, B2B, and complex enterprise hierarchies
- CRM does not define products independently (Catalog driven)
- Central billing operations with local compliance
- Convergent Billing System to combine charges from various services into a single bill for end-user
- The legacy/global backbone for billing
- Cerillion Billing is the end-to-end commercial billing system that manages order-to-cash, invoicing, and customer billing lifecycle

Oracle BRM (2023-24)

- New billing platform for expansion
- Oracle BRM powers new digital growth domains
- Includes Oracle Pricing Design Center as a new, production billing and revenue management platform to support its expanding digital, B2B2X, and white-label connectivity offerings
- Oracle BRM is the central monetization and revenue management platform that converts usage into accurate, auditable charges and balances

Oracle Pricing Design Center(2023-24)

- New Product Catalog for select domains
- Configures real-time pricing(usage-based charging), offline pricing (recurring, one-off charges), tiered pricing
- Supports mass-market products, enterprise contracts, partner-specific commercial models

Digital & Experience Layers (2023-24)

- Microservice-oriented design
- Containerized workloads on Kubernetes (EKS)
- AWS as a core platform
- 1global connect open API

https://www.optiva.com/whitepapers-reports-open/ocs-on-public-cloud-transforms-1global's-business-inside-and-out
1Global: Multi-country Billing Case Study | Cerillion
https://www.cerillion.com/products/bssoss-suite/crm-plus/
https://bebee.com/pt/jobs/oracle-brm-developer-lisbon-based-1global-lisbon--fj-2018713721

https://docs.connect.1global.com/docs/overview/whatisconnect
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https://builtin.com/job/oracle-brm-developer/8255389

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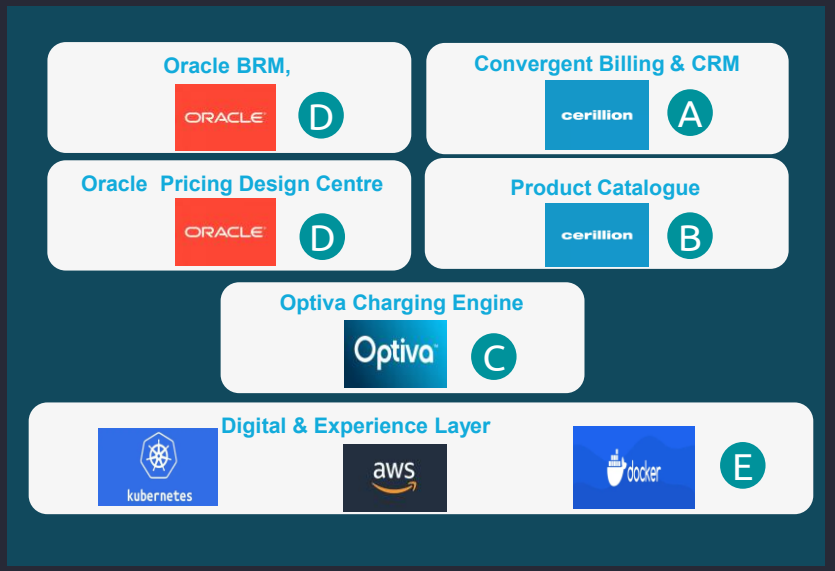
1global overview

1global consolidates core IT platforms, services, and capabilities at enterprise scale to drive efficiency, consistency, and value, while enabling businesses to focus on growth within clear governance guardrails. Accountability is maintained through clearly defined federated model, decision rights, and KPIs driven by business outcome.

Context & Scope

- Transforming IT across 12 countries and has a network which spans across 190+ countries across the globe.
- Transformation to accommodate multiple tenancy structure.
- Cross segment mutualization with outcome-based accountability
- Centralized security & data standards
- Joint roll out and benefit realization
- Implement a cloud-native, API-driven design
- Localized and adaptive feature prioritization and change approach

Key Solutions



- A Global Backbone for Billing
- B One Global Catalog Instance
- C Real-time Online Charging System
- D Central Monetization & Revenue management platform
- E Cloud Native

Operating Model & Approach

- 1GLOBAL runs on global telecom core serving multiple markets, customers, and products; through shared infrastructure and services
- Smart network switching for fail-safe connectivity
- Implemented a tightly integrated Catalog-Billing-Charging model to support multi-country business model
- Near real-time usage tracking, alerts and rules engine
- Cloud native horizontally scalable infrastructure design
- Rapid partner onboarding through APIs
- A federated governance model with centralised control while preserving local autonomy

Learnings

- Cloud-native BSS stack transformations achieve greater agility and multi-tenancy capability
- Use centralized portals and APIs for real-time control
- Offer automated provisioning and service lifecycle management
- KPI's driven by business outcome is more effective compared to metrics-based on IT Systems

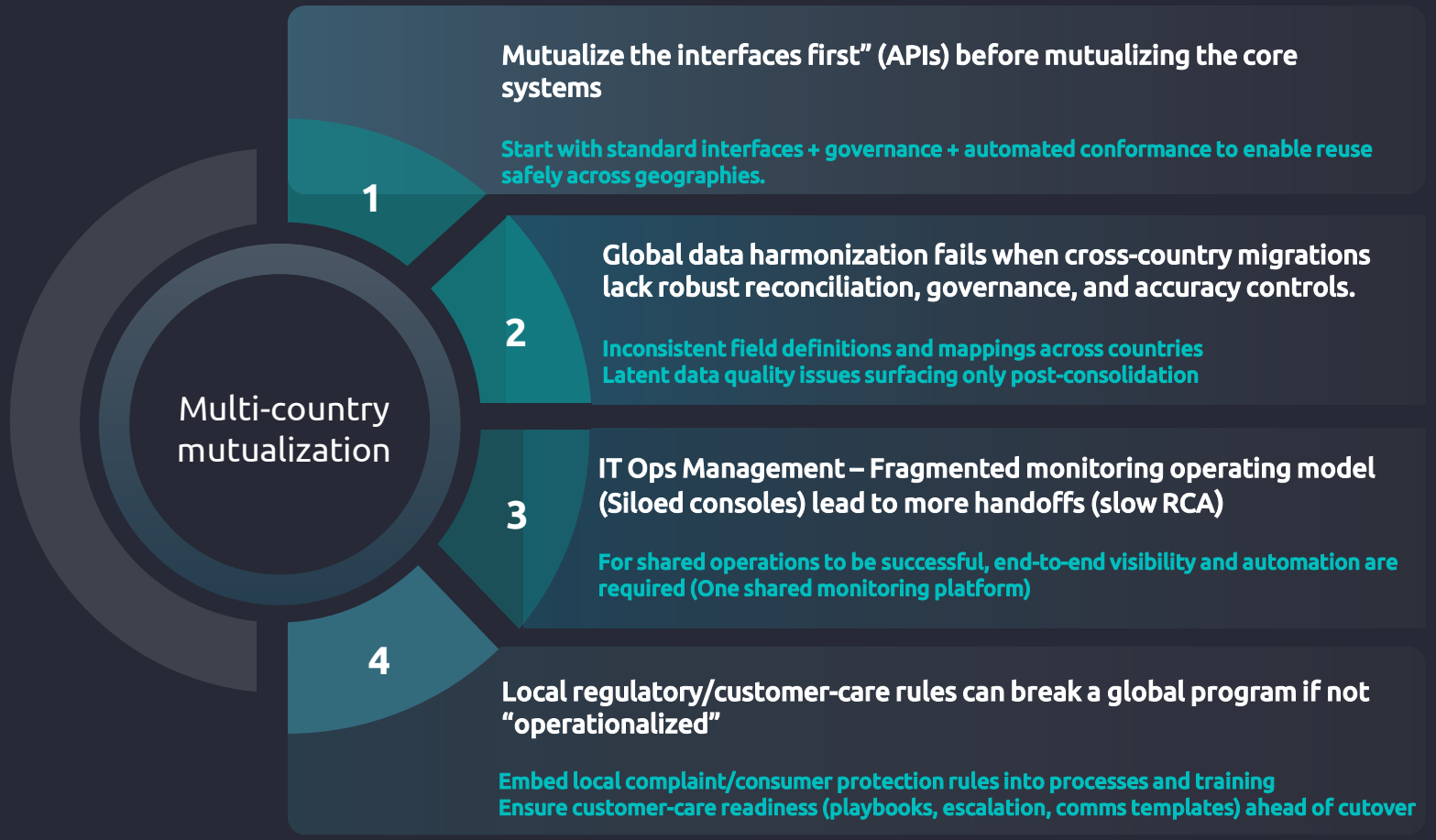
Results: Implement Unified Product Catalog

- Time to Market: 30-60% ↑
- Product Build Effort: 50-70% ↓
- BSS customization cost: 20-40% ↓
- Product variants across OpCos: 30-50% ↓
- Defects & Fallout: 40-60% ↓
- Re-use Rate: 2-4 x per product component ↑

IoT Connectivity Solutions | Connectivity Management Platform
1GLOBAL Use Cases across Industries

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Learnings from failure - Vodafone



FAILURE PATTERN CASE STUDY

Vodafone's IT mutualisation failure occurred in the UK during CRM and billing consolidation onto Oracle Siebel, resulting in billing errors for over 10,000 customers and a £4.6M regulatory fine

Data migration + Controls + Detection Gap

Scale and complexity of migration (small proportion of customer accounts were incorrectly migrated). Mistakes in billing data and price plan records in the new system

Vodafone did not act quickly enough, Ofcom notes that effective steps happened after intervention (operational reconciliation (detect → alert → fix → contain) did not work effectively)

Customer service processes were not "implementation-ready" for the failure modes (complaints definition, escalation, regulatory communications)

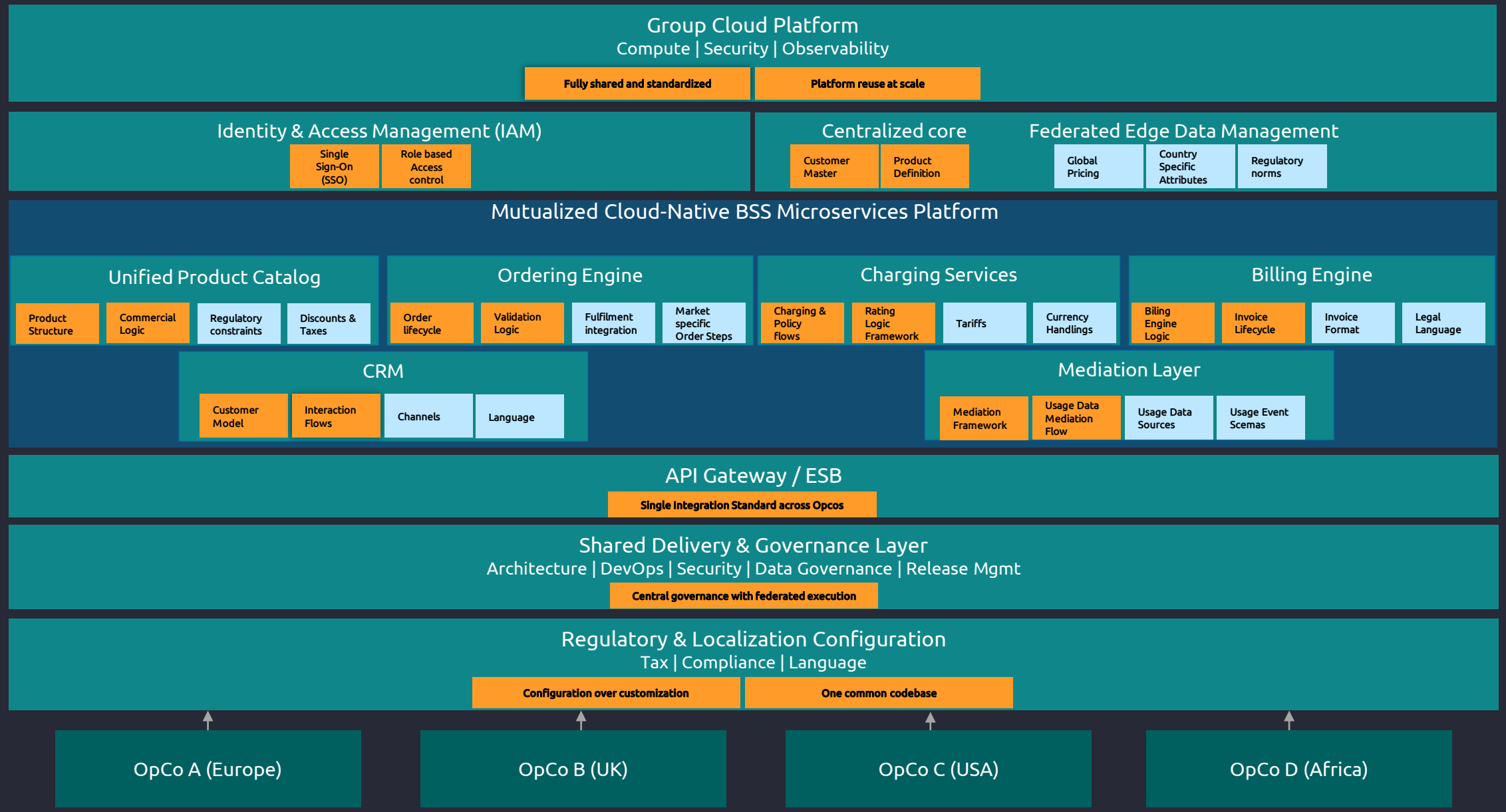
Missing automated anomaly detection and reconciliation controls for prepaid top-ups created a blind spot where funds could be collected without being reflected in customer balances

Future Roadmap - Vodafone continued its mutualization agenda - shifting toward modularization, standard interfaces (Open APIs), stronger governance and shared-services operating models rather than "big-bang" platform swaps everywhere

https://inform.tmforum.org/features-and-opinion/my-api-story-campaign-fostering-cohesive-global-digital-experiences
https://info.tmforum.org/Automating-API-conformance-cuts-Vodafones-integration-costs-by-multiple-mi…html
https://www.bbc.com/news/business-37772118

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Recommended IT Mutualization Architecture



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Best Practices of Mutualization

Drawing on TM Forum's Catalyst outputs, operator case studies, and industry analyst frameworks, the following represent the most critical positive practices for BSS mutualization programmes.

Design a Phased Roadmap with Explicit Value Gates

Structure the programme as a series of phases, each with a defined business value gate that must be achieved before the next phase is funded and initiated.

A typical structure:

- Phase 1 - shared product catalogue and unified customer master
- Phase 2 - shared order management and provisioning
- Phase 3 - converged billing and revenue management

Invest Heavily and Early in Data Governance and Migration

- Treat data migration as a first-class workstream - not a sub-task of system integration. Appoint a dedicated Chief Data Officer or Data Migration Lead with programme-level authority.
- Begin data profiling, cleansing, and harmonization activities at programme inception, in parallel with architecture design. Establish a canonical data model aligned to TM Forum SID (Shared Information & Data Model) as the target state for all customer, product, and billing data.
- Budget at minimum 20-25% of total programme cost for data-related activities - operators who budget less consistently overrun.

Establish a Strong, Empowered PMO with Cross-Entity Authority

- A Programme Management Office with genuine cross-entity authority - staffed by senior representatives from each participating BU or OpCo, reporting directly to the Group CTO or CIO - is essential.
- The PMO must have authority to resolve disputes, enforce architectural standards, and escalate blockers to executive level.
- Critically, PMO leadership should include both IT and business representatives with equal standing; a purely IT-led PMO will struggle to maintain business alignment and executive engagement over a multi-year programme horizon.

Anchor the Entire Programme on TM Forum ODA and Open APIs

- Mandating TM Forum ODA alignment and Open API conformance from day one - as a contractual requirement for all vendors, not an aspiration - provides the technical discipline that prevents mutualization from recreating the same integration complexity it was designed to eliminate
- Require vendors to provide TM Forum conformance certification for all components.
- Establish an API governance board with authority to reject non-compliant integration proposals.
- Document all integration points in a shared API catalogue accessible to all programme participants.

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Backup Slides

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IT Stack Mutualization | Critical Questions for Mutualization

Assessment Category	Questions Need to Addresses to Embark on Mutualization Journey
1 Strategic Assessment	1.1 Application wise Complexity, Business Criticality, Frequency of Change, serving business segment (Consumer/ SME/ Enterprise), serving product line (Fixed/ Mobile/ Converged etc.) Obsolescence etc. 1.2 Is the long-term ambition for partial reuse, full common stack, or shared services across all markets?
2 Methodology & Vendor Assessment	2.1 Which Vendors are currently delivering services (mapped to either domain or application or market segment). Will there be a challenge in terms of vendor support for a new market mutualization? 2.2 Design methodology and architecture conformance process complicated to impact mutualization
3 IT Stack Assessment & Enablement	3.1 Architecture Roadmap, Runway, status of Tech Debt, Applications Performance impacting Mutualization 3.2 Is the IT stack & its components COTS Technology or homegrown? How scalable and compact is the stack? Is the current stack modular and API driven to be taken across markets
4 Lift & Shift Feasibility & Obsolescence Assessment	4.1 Are these Assets developed for Reuse, How is standardization and reuse adopted in the stack? 4.2 Are there any legacy systems stack that were acquired and are still running? Are there any consolidation plans of these legacy stacks?
5 Cost & Risk Assessment for Mutualization	5.1 What are the cost synergies from consolidating platforms? 5.2 Can the shared stack enforce zero-trust and security-by-design across all markets?

MUTUALIZATION DECISION FRAMEWORK | OVERALL STACK LEVEL

Telcos expanding into new markets face high upfront IT investments & long deployment cycles with mutualizing an existing, proven IT stack accelerates entry and reduces duplication

EXPLORE TO MUTUALIZE

Identify and extract IT Stack components from the current Markets landscape to Mutualize.

DESIGNED TO MUTUALIZE

Identify and explore Operating Model effectiveness across People, Process & Application for Mutualization

DEFINED TO MUTUALIZE

Assess the IT stack capabilities from reuse, duplicate or expand in terms of customizations, APIs etc.

The Key Considerations that drive the tenets of Mutualization

	Key Tenets	Area of Consideration for Decision Making
1	**Mutualization Scope Definition & Design** (What will be shared, Standardized, Centralized)	IT Platforms, Network Assets, Channels, Cybersecurity, Analytics Model, Shared Infra, Product Catalogue
2	**Operating Model Assessment & Design** (How will the mutualized environment run)	Federated vs Centralized vs Distributed with Governance structure
3	**Technology Architecture Assessment & Design** (Is the technology stack ready for mutualization)	Multi Tenant, Multi Country Cloud Infrastructure, Nativity, standardized API driven workflows, Security consideration
4	**Data Risk & Sovereignty Management** (How will data platform look like)	Data Flows & Management as per security guidelines, GDPR, ICU and hosting, Shared Risks, Dependencies
5	**Risk Assessment & Governance** (What are the key risks & mitigation)	Regulatory compliance risk mitigation, Cyber Security Risk mitigation, Business Continuity Planning
6	**Financial Decision Effectiveness** (Does it create financial value to mutualize)	Capex Avoidance, Opex Reduction with well defined ROI timeline (IT Ops, Licenses, Infra cost, Vendor mgmt. etc.)
7	**Change Management & Absorption Enablement** (what are key activities for change)	Shared Service Model with local adoption program, SOPs, Playbooks creation, KM Transfer, Language Barriers etc.

Mutualization Decision Assessment

Decision 1: Cross Market Stack Presence (High)

Asset deployed currently in multiple markets easier to mutualize



Decision 2: Cross Segment Mutualization (High)

Asset developed in a market segment is being reused across other relevant segments



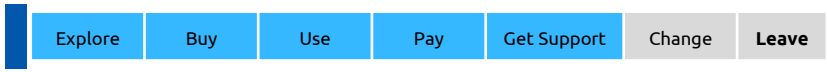
Decision 3: Cross Product Mutualization (High)

Asset developed for a Product is reused across other relevant Products



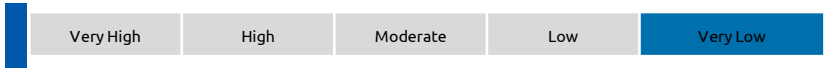
Decision 4: Cross Journey Mutualization (High)

Asset developed for a Journey is reused across other relevant Journeys



Decision 5: Risk Profiling for Mutualization (Very Low)

Overall Risk associated with Stack Mutualization



IT Stack Mutualization | Evaluation Framework at Application Level

Technical Levers for Mutualization Profiling

Complexity Criticality Change Frequency Customisations
Obsolescence Presence in markets Cloud Nativity
Customer Experience with Stack/App

Business Levers for Mutualization Profiling

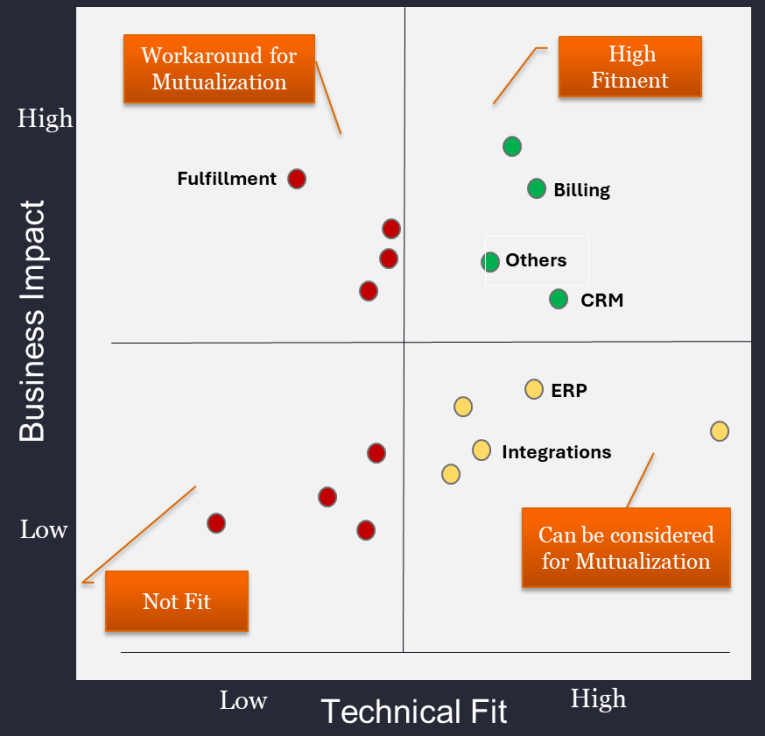
Cost Organization Capability & Roadmap Standards
Business Impact Scalability Factor Agility
Security Risks Data Risks

Mutualization Evaluation - BSS Stack

	Complexity	Criticality	Customization	Obsolescence	Markets Presence	Cloud Nativity
Billing						Mix of Cloud & On-premise
CRM						Mix of Cloud & On-premise & Virtualized
ERP						Mix of Cloud & Virtualized
Fulfillment						Mostly Virtualized
Integrations						Mostly Virtualized
Others						Mostly Virtualized

Very High High Moderate Low Very Low

Mutualization Mapping



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